

# A Short Report on Simulation Based Inference

Scientific Machine Learning Seminar

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## Abstract

Simulation Based Inference (SBI) offers a novel approach to solve inverse problems powered by machine learning. Hereby, a neural network learns to relate datapoints from a simulated dataset to a posterior distribution of model parameters. It can outperform classical Bayesian inference methods in inference speed and posterior accuracy. Additionally, SBI does not require any likelihood evaluation which often is computationally costly or has to be approximated. Common problems like model misspecification, neural inaccuracies or miscalibrated posteriors can limit the usefulness of SBI if they are not circumvented. Exemplary usecases for SBI are inferring the complex parameter posteriors from planetary microlensing events or gravitational waves from black hole mergers.

## 1 Introduction

In computational physics there are usually two types of problems to solve: Firstly, forward problems where the parameters and/or boundary conditions of a model are given and the goal is to find data or a data distribution which adheres to this model. For example in the case of given parameters and initial values this is a typical simulation.

Though secondly, there are also inverse problems where the data is the preset and now the parameters or parameter distributions of the assumed model have to be inferred. So here the data likely comes from an observation and the parameters which represent physical properties of the system have to be computed. This is the type of problem that will be explored in this report.

A rather simple case of solving an inverse problem is the use of linear regression to fit a function to a given dataset. But for more complex models also methods with higher efficiency are needed. The classical approach here is to use Bayesian methods to infer the parameter posterior distribution, but Simulation Based Inference introduces machine learning techniques to solving inverse problems. SBI is able to solve these problems in some cases more efficiently and computationally cheaper. Applications where SBI shows advantages are (but not limited to) gravitational microlensing (Zhang et al., 2021), gravitational wave physics (Dax et al., 2023) and neuroscience (Deistler et al., 2025, Sec. 4.2).

In Section 2 the general concepts of classical Bayesian inference are first explored, specifically focussing on Markov-Chain-Monte-Carlo methods. Then SBI and its underlying concepts are introduced in Section 3 and exemplified in Section 4 with the help of a toy-example. The advantages and limitations of SBI are presented in Section 5. Two exemplary applications with

results are portrayed in Section 6 and a conclusion is given in Section 7.

## 2 Classical Bayesian Inference

General concept of Bayes equation

$$p(\theta|x) = \frac{p(x|\theta)}{p(x)}p(\theta) \tag{1}$$

Methods of inference, MCMC

Limitations: convergence, local approximation, multimodal posteriors

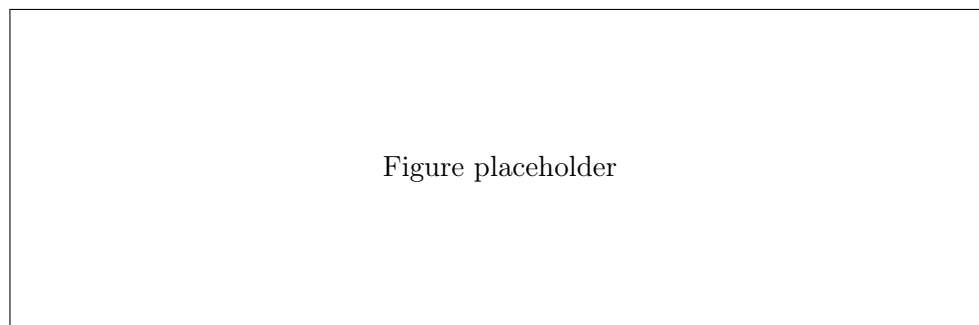
## 3 Introduction to Simulation Based Inference

Key concepts:

Skip Likelihood inference, implicit in simulation

Neural Posterior Estimation (NPE)

Sampling from simulator, train neural network, estimate posterior from observation Fig. 1



**Figure 1:** Workflow of SBI.

## 4 Simple SBI Example

Example: angled throw

choose parameter prior, create simulator of throw, simulate dataset

train neural network, here normalizing flows, any network works

loss function

$$L(\phi) = \tag{2}$$

use network to evaluate observation, data and param interval

## 5 Benefits and Limitations

Key benefits: very fast inference (Amortized Inference), Likelihood-free (no approximation or expensive computation), multimodal posteriors

Limitations: Upfront training, model misspecification (common pitfall) posterior is a neural estimation uncalibrated posterior

## 6 Application

Short explanation of microlensing

Results of usage of SBI ([Zhang et al., 2021](#))

Short explanation of gravitational waves and black hole mergers

use of flow matching networks

results of usage here ([Dax et al., 2023](#))

Other applications

## 7 Conclusion

Benefits and Limitations (stuff from abstract)

When to use SBI inverse problem, time critical inference, expensive likelihood evaluation, multimodal posterior

## References

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